Cody Freitag

codyrfreitag@gmail.com codyrfreitag.me

Professional Experience

Cornell Tech, *Postdoctoral Researcher* Oct. 2026–Present

Columbia University, *Adjunct Assistant Professor of CS* Spring 2027

- Appointed to teach Algorithms in the Department of Computer Science.

Proof Trading, *Research Consultant* Oct. 2026–Present

- Developing mathematical models for analyzing information leakage in algorithmic trading.

Previous Appointments

Hebrew University of Jerusalem, *Postdoctoral Researcher* 2025–2026
Northeastern University, *Postdoctoral Researcher* 2023–2026
Boston University, *Postdoctoral Researcher* Summer 2023

Selected Research

Cryptographic Proofs
Developed new techniques for succinct cryptographic proofs that overcome asymptotic bottlenecks in prover running time and space complexity.

Decentralized Computation
Designed time-based cryptographic primitives and protocols for decentralized randomness generation and sealed-bid auctions.

Foundations of Cryptography
Established constructions and barriers for unique cryptographic proofs, with implications for the computational hardness of finding Nash equilibria.

Modeling and Analyzing Real-World Systems
Developed mathematical models for analyzing privacy and information flow, ranging from statistical testing to information leakage in algorithmic trading systems.

Publications: Research has appeared at STOC, Crypto, and Eurocrypt, among other venues. full list

Education

Cornell University 2023
Ph.D. and M.S. in Computer Science.

The University of Texas at Austin 2017
B.S. in Computer Science; B.S. in Mathematics.

Additional Experience

- **Industry and software engineering:** Developed software at Bloomberg, L-3 Communications, and Zeutro; consulted on cryptographic technologies for Yeletech Security and Bolt Labs.
- **Teaching and communication:** Presented research at international conferences and university seminars; served as a teaching assistant for undergraduate and graduate courses in algorithms, discrete mathematics, and related subjects.